

Intraoperative Absolute Depth Estimation in MVD Surgery

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Abstract—Microvascular decompression (MVD) is a neurosurgical procedure that relieves nerve compression by repositioning or separating offending blood vessels, effectively reducing pain or spasms. Accurate localization of the compression site is crucial for optimal surgical outcomes, as it enables precise identification and decompression of the offending vessel. While horizontal anatomical relationships are easily identified in the surgical view, compressions occurring along the depth axis are more challenging to discern. In this study, we propose a method to measure accurate intraoperative distances during MVD surgery using Depth-Anything-V2. By leveraging the optical properties of standard imaging equipment in conjunction with the depth estimation model, our method computes precise, absolute distances rather than relying solely on relative measurements, achieving distance estimation errors of less than 2 mm compared to intraoperative and preoperative reference measurements.

Index Terms—Deep Learning, Depth Estimation, Surgical Video Analysis, Microvascular Decompression (MVD)

I. INTRODUCTION

Microvascular decompression (MVD) is a neurosurgical procedure to relieve nerve compression by repositioning blood vessels. Accurate localization of the compression site is critical for surgical success, but complex anatomy and limited intraoperative visualization make this challenging [1]. Current surgical procedures use preoperative MRI or CT for planning, with intraoperative anatomy assessed via endoscopic or microscopic observation. However, these views are typically two-dimensional and lack depth information, limiting spatial understanding. Fig. 1 shows the spatial relationships of key anatomical structures for MVD.

Recent studies have explored deep learning-based monocular depth estimation for enhancing surgical navigation without requiring stereo hardware [2], [3]. However, these methods produce only relative depth and lack real-world scale, which is crucial for MVD. In this work, we propose a method to estimate absolute depth by combining monocular depth prediction with optical characteristics of surgical microscopes.

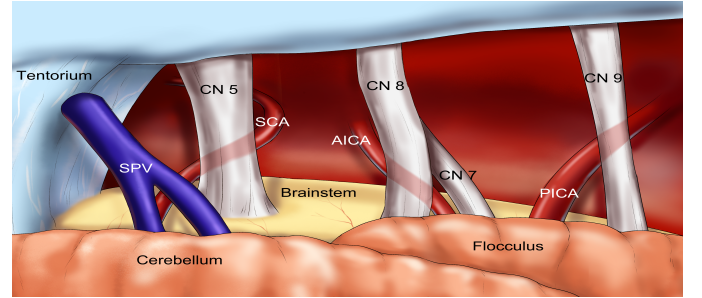


Fig. 1. Schematic of neurovascular anatomy in the cerebellopontine angle, showing key structures and their relationships during MVD.

Using focal adjustments, we extract relative shifts and calibrate them against predicted depth maps to recover metric distances. Our approach is validated using preoperative MRI-based 3D reconstructions and demonstrates potential for intraoperative application.

II. METHODS

The overall pipeline is shown in Fig. 2. Intraoperative depth estimation is performed using a monocular model to obtain relative depth, which is then calibrated to absolute depth using optical measurements from a ZEISS TIVATO 700 surgical microscope [4]. The resulting absolute distances are validated against preoperative 3D MRI reconstructions.

A. Relative Depth Estimation

We use the pre-trained Depth-Anything-V2 [3] to estimate relative depth in the surgical field. To enhance accuracy, we focus predictions on the region of interest (ROI) around critical neurovascular structures.

B. Absolute Depth Calibration via Optical Scaling

Absolute depth is calibrated using a surgical microscope, which provides focal length readouts and allows precise focus adjustments. The physical distance between known anatomical

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TABLE I
REPRESENTATIVE MEASUREMENTS FROM SELECTED PATIENTS. MEAN ERRORS ARE CALCULATED FROM ALL AVAILABLE CASES.

Subject	Target Structures	Measured (mm)		Estimated(mm)	Errors (mm)	
		Microscope	3D-Slicer		Microscope	3D-Slicer
Patient 1	CN8–CN6	16.0	17.37	15.97	0.03	1.4
Patient 2	CN8–CN9	1.0	1.954	1.71	0.71	0.244
Patient 5	CN5–CN9	10.0	7.829	8.39	1.61	0.561
Patient 8	CN5–vein	-	7.781	10.75	-	2.969
Patient 10	CN5–CN9	13.0	13.23	5.73	7.27	7.5
Patient 11	CN5–artery	-	6.154	4.19	-	1.964
Average (all cases)		-	-	-	3.197	3.326
Average (w/o extreme cases)		-	-	-	1.504	1.897

landmarks is measured using the microscope’s focal adjustment, establishing a scale factor to convert relative depth into real-world distances.

C. Validation of Absolute Depth Using 3D Slicer

To validate the absolute depth measurements, we used 3D Slicer, a platform for medical image analysis and 3D reconstruction [5]. Volumetric models were generated from preoperative MRI scans, providing reference distances for evaluating intraoperative depth. We manually aligned the surgical scene captured by the microscope with the corresponding 3D model and compared distances between anatomical structures.

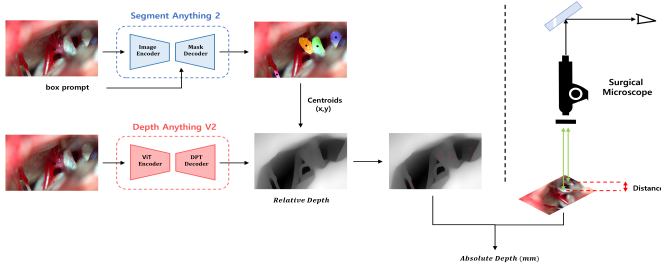


Fig. 2. Overview of the pipeline. Relative depth maps are generated using Depth-Anything-V2 [3], with anatomical structures segmented by Segment-Anything-Model 2 [6]. Absolute distances are then computed by calibrating depth differences with intraoperative microscope measurements.

III. RESULTS AND DISCUSSION

We compared absolute distances between anatomical landmarks derived from predicted depth maps with reference distances from 3D Slicer reconstructions. Fig. 3 presents side-by-side comparisons of surgical images, depth maps, and aligned 3D views with annotated measurements. As summarized in Table I, our estimates closely match reference distances, with an average error of 3 mm. When excluding outliers, this error drops to 1.8 mm, demonstrating strong accuracy and reliability.

We present a method for estimating absolute depth during MVD surgery by calibrating monocular depth predictions using intraoperative optical measurements. This enables real-time distance estimation between anatomical structures from 2D surgical images, with an accuracy of around 2mm compared to 3D Slicer references. This indicates that 3D Slicer reconstructions can reliably replace real-world depth measurements intraoperatively. As shown in Figure 4, while effective

in most cases, the method struggles in scenes with large depth variation, often underestimating distances to farther structures. Future work includes improving calibration accuracy and using depth-sensing microscopes to develop intraoperative depth models for real-time 3D reconstruction.

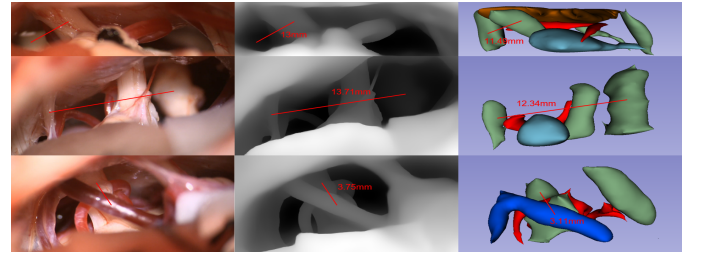


Fig. 3. Comparison of absolute depth measurements from intraoperative images, Depth-Anything-V2 predictions, and 3D Slicer reconstructions, with annotations indicating inter-structural distances.

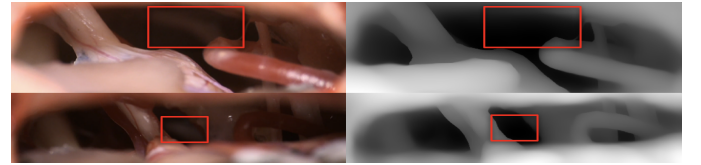


Fig. 4. Depth information is noticeably reduced for deep structures, indicating the model’s limited ability to preserve distant depth variations.

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